Tatiana Zueva

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Summary

Software Engineer focused on real-time 3D systems and computer vision. Built XR applications and 3D reconstruction pipelines in C++, with experience across Unity, Unreal Engine, Python, and C#.

Skills

XR & 3D: Unity, Unreal Engine, OpenGL, GLSL, WebXR, CloudXR
Computer Vision & AI: OpenCV, Dlib, FaceDetection, FaceTracking, 3D reconstruction, Gaussian Splatting
Languages: C++, C#, Python, JavaScript

Experience

Software Engineer (XR / 3D / Computer Vision), Technical Lead *03/2024 — 04/2025*
MTS Web Services (Future Crew) *— R&D innovation unit*

- Led end-to-end development of an XR platform for real-time 3D reconstruction and visualization
- Designed and implemented a Gaussian Splatting pipeline enabling real-time 3D rendering *(*.ply, *npz, *.splat)*
- Optimized rendering and data processing pipelines to achieve real-time performance in XR environments
- Developed cross-platform XR applications using Unity (C#) and WebXR (three.js, React)
- Defined system architecture for scalable real-time 3D and XR systems
- Built, led, and managed an 8-person cross-functional team (VR, ML, DevOps, QA, Analytics)
- Owned technical roadmap and execution in collaboration with executive stakeholders

Technologies: C#, Python, Unity, WebXR, three.js, CloudXR
3D / AI: Gaussian Splatting, 3D reconstruction

Senior Software Engineer *07/2022 — 12/2023*
Avatai *— XR startup*

- Led development of a VR application for volumetric video playback (Oculus Quest, HTC Vive) as a hands-on engineer
- Drove the project from early development to public release on Steam (Spherum 3D)
- Implemented core application logic and playback systems in Unity, focusing on performance and UX
- Developed reusable cross-platform Unity components for VR and mobile applications
- Designed and built cloud infrastructure (AWS, Firebase) for scalable asset delivery
- Implemented CI/CD pipelines and automated build and release workflows

Software Engineer *10/2021 — 06/2022*
Avatai *— XR startup*

- Developed a VR application for volumetric video playback (Oculus Quest, HTC Vive) using Unity
- Implemented client-side API integration for social features
- Reduced application build size using Unity Addressables and Firebase-hosted asset bundles

Software Engineer *06/2021 — 10/2021*
CROC Immersive Technologies *— VR division of a leading IT systems integrator*

- Developed a VR training system for Dubai Municipality, implementing game mechanics and interactive learning scenarios
- Customized text component in Unity for localization using editor scripting
- Designed and implemented inner development tools to speed up development iterations
- Developed an automated level generation system using procedural algorithms to create dynamic and varied game environments

Software Engineer *12/2014 — 03/2017*
NTI Center *— university-based XR R&D center*

- Led the design and development of a VR-based language learning application using Unreal Engine and AI technologies.
- Built immersive learning scenarios with interactive NPCs
- Integrated Microsoft Cognitive Services for real-time speech recognition and dialogue
- Enabled bidirectional communication between users and virtual characters

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Education

Far Eastern Federal University (FEFU), Russia
M.S. Computer Science (Game Development, VR/AR), 2017–2019 — **GPA: 5.0/5.0**
B.S. Computer Science (Systems Programming), 2013–2017 — **GPA: 4.6/5.0**

Projects & Achievements

Coins Recognition at Sberbank — Real-time Object Detection & Classification

- Contributed to the development and optimization of a real-time coin detection and classification system using YOLOv3 (Darknet)
- Worked with a large-scale dataset (>400,000 annotated images across ~200 classes) using CVAT and imgaug
- Achieved >85% detection accuracy with low-latency inference suitable for real-time production use
- Demonstrated the system at the Eastern Economic Forum (Vladivostok, 2019)

Technologies: Python, Darknet (YOLOv3), OpenCV (FaceDetection, SolvePnP, FaceTracking), CVAT, imgaug

Hairstyle AR — Real-time Face Detection & Tracking (C++ / AR)

- Designed and implemented a real-time AR application for interactive 3D hairstyle try-on, combining computer vision and 3D rendering.
- Built a face detection and tracking pipeline using OpenCV and Dlib
- Implemented real-time 3D rendering and overlay using OpenGL
- Developed application logic and UI using Qt (C++)
- Optimized processing pipeline for low-latency AR interaction

Open-source project: AR-Hair (15 stars, 7 forks)
Technologies: C++, Qt, OpenCV, Dlib, OpenGL

MATE ROV 2017 (International Finals, Los Angeles) — Remote Operated Vehicle Control System

- Designed and implemented a control interface for an underwater robotic system using Qt (C++), focusing on real-time interaction and operator control.
- Built interactive UI for real-time control and system feedback
- Implemented input handling and control logic for robotic operation
- Developed network-based communication (UDP) for real-time control
- Worked with camera/video streams using Qt Multimedia

Links: [Technical Report] | [Spec Sheet]
Technologies: C++, Qt, Qt Multimedia, SDL-2, UDP networking

Competitions & Certifications

- **Prize-Winner, ACM ICPC (2015/16 & 2016/17, Quarterfinals)** — advanced algorithms and problem-solving skills.
- **Certified Expert, IT Software Solutions for Business — WorldSkills (2018–2020)**.
- **Author & Jury Member, VR Track, National Technology Olympiad, Russia (2019–2025)** — designed challenges, mentored participants, evaluated VR projects.